

MATH 129: Big Data Homework 1

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1) Linear transformation

Let $y = Ax + b$ be a random vector. Show that the expectation is linear:

$$a) \quad \mathbb{E}[y] = \mathbb{E}[Ax + b] = A\mathbb{E}[x] + b.$$

Also show that

$$b) \quad \text{cov}[y] = \text{cov}[Ax + b] = A \text{cov}[x] A^T = A \Sigma A^T$$

a) Proof:

We are given:

$$y = Ax + b$$

Take the expectation of both sides:

$$\mathbb{E}[y] = \mathbb{E}[Ax + b]$$

By linearity of expectation, expectation distributes over addition:

$$\mathbb{E}[y] = \mathbb{E}[Ax] + \mathbb{E}[b].$$

Because A is a deterministic matrix, we can pull it outside the expectation:

$$\mathbb{E}[Ax] = A\mathbb{E}[x].$$

Similarly, because b is a deterministic vector:

$$\mathbb{E}[b] = b$$

Substituting these results:

$$\mathbb{E}[y] = A\mathbb{E}[x] + b$$

Therefore,

$$\mathbb{E}[Ax + b] = A\mathbb{E}[x] + b$$

$$b) \text{ covariance } \text{cov}[y] = \text{cov}[Ax + b] = A \text{cov}[x] A^T = A \Sigma A^T$$

Proof:

Recall the definition of covariance:

For a random vector x , its covariance matrix is:

$$\text{cov}(x) = \mathbb{E}[(x - \mathbb{E}[x])(x - \mathbb{E}[x])^T]$$

Let $\mu = \mathbb{E}[x]$. Then:

$$\Sigma = \mathbb{E}[(x - \mu)(x - \mu)^T]$$

We can apply this definition to y :

$$\text{cov}(y) = \mathbb{E}[(y - \mathbb{E}[y])(y - \mathbb{E}[y])^T]$$

We know that:

$$y = Ax + b$$

and from part a):

$$\mathbb{E}[y] = A\mathbb{E}[x] + b$$

Substituting both expressions gives:

$$\text{cov}(y) = \mathbb{E}[(Ax + b - A\mathbb{E}[x] - b) \cdot (Ax + b - A\mathbb{E}[x] - b)^T]$$

Notice: $+b - b = 0$. So,

$$= \mathbb{E}[(Ax - A\mathbb{E}[x]) \cdot (Ax - A\mathbb{E}[x])^T]$$

Pull common term A out:

$$= \mathbb{E}[A(x - \mathbb{E}[x]) (x - \mathbb{E}[x])^T A^T]$$

Recall the transpose rule:

$$(AB)^T = B^T A^T$$

So,

$$[A(x - \mathbb{E}[x])]^T = (x - \mathbb{E}[x])^T A^T$$

Substituting:

$$\text{cov}(y) = \mathbb{E}[A(x - \mathbb{E}[x]) (x - \mathbb{E}[x])^T A^T]$$

Because A and A^T are deterministic:

$$\text{cov}(y) = A \mathbb{E}[(x - \mathbb{E}[x]) (x - \mathbb{E}[x])^T] A^T$$

The expression within the matrix is precisely the definition of Σ :

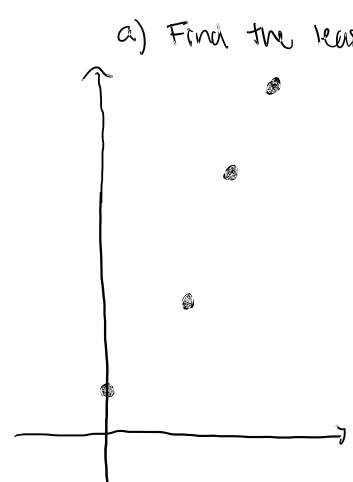
$$\mathbb{E}[(x - \mathbb{E}[x]) (x - \mathbb{E}[x])^T] = \Sigma$$

Thus,

$$\text{cov}(y) = A \Sigma A^T.$$

$$2) \text{ Given the dataset } D = \{(x, y)\} = \{(0, 1), (2, 3), (3, 6), (4, 8)\}$$

a) Find the least squares estimate $y = \theta^T x$ by hand using Cramer's Rule.



$$D = \{(x, y)\} = \{(0, 1), (2, 3), (3, 6), (4, 8)\}$$

$$\tilde{y} = \theta^T x$$

$$\leftarrow \text{linear: } y = mx + b$$

$$\text{prediction error: } e_i = y_i - (mx_i + b)$$

least squares minimizes the sum of squared errors:

$$J(m, b) = \sum_{i=1}^4 (y_i - mx_i - b)^2$$

Substitute 4 observations:

$$(0, 1): J(m, b) = (1 - 0m - b)^2 = (1 - b)^2$$

$$(2, 3): J(m, b) = (3 - 2m - b)^2$$

$$(3, 6): J(m, b) = (6 - 3m - b)^2$$

$$(4, 8): J(m, b) = (8 - 4m - b)^2$$

$$J(m, b) = \sum_{i=1}^4 (y_i - mx_i - b)^2$$

$$J(m, b) = \sum_{i=1}^4 (y_i - mx_i - b)^2$$

$$u(b) = y_i - mx_i - b$$

$$\frac{\partial}{\partial m} [u(m)]^2 = 2 u(m) \cdot u'(m)$$

$$\frac{\partial u}{\partial b} = 0 - 0 - 1 = -1$$

$$u'(m) = y_i - mx_i - b$$

$$\frac{\partial}{\partial m} = 0 - x - 0$$

$$\frac{\partial J}{\partial b} = -2 \sum_{i=1}^4 (y_i - mx_i - b)$$

$$\frac{\partial J}{\partial m} = -2 \sum_{i=1}^4 x_i (y_i - mx_i - b)$$

$$\sum_{i=1}^4 (y_i - mx_i - b) = 0$$

$$\sum_{i=1}^4 x_i (y_i - mx_i - b) = 0$$

$$\sum y_i - m \sum x_i - 4b = 0$$

$$\sum x_i y_i - m \sum x_i^2 - b \sum x_i = 0$$

$$\sum y_i = m \sum x_i + 4b$$

$$\sum x_i y_i = m \sum x_i^2 + b \sum x_i$$

two normal equations:

$$\sum y_i = m \sum x_i + 4b$$

$$\sum x_i y_i = m \sum x_i^2 + b \sum x_i$$

$$\begin{array}{cc|cc} x_i & y_i & x_i^2 & x_i y_i \\ 0 & 1 & 0 & 0 \\ 2 & 3 & 4 & 6 \\ 3 & 6 & 9 & 18 \\ 4 & 8 & 16 & 32 \\ \hline \text{sums:} & 9 & 29 & 56 \end{array}$$

$$18 = 9m + 4b$$

$$56 = 29m + 9b$$

$$\begin{bmatrix} 9 & 4 \\ 29 & 9 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} 18 \\ 56 \end{bmatrix}$$

$$\left. \begin{array}{l} \sum x_i = 9 \\ \sum x_i^2 = 29 \\ \sum x_i y_i = 56 \end{array} \right\} \begin{array}{l} \text{a system of two equations} \\ \text{w/ 2 unknowns} \end{array}$$

Cramer's Rule: For a system: $M\theta = c$. Cramer's rule states:

$$\theta_j = \frac{\det(M_j)}{\det(M)}, \text{ where } M_j \text{ is obtained by replacing the } j\text{th column of } M \text{ with } c$$

$$\text{result: } \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

$$M = \det \begin{bmatrix} 9 & 4 \\ 29 & 9 \end{bmatrix} = 4(29) - 9(9)$$

$$= 116 - 81 = 35 \leftarrow \text{nonzero determinant, system has a unique solution}$$

Solve for intercept b :

$$M_b = \begin{bmatrix} 18 & 9 \\ 56 & 29 \end{bmatrix} \quad \det(M_b) = 18(29) - 9(56) = 522 - 504 = 18$$

$$\text{apply Cramer's rule: } b = \frac{\det(M_b)}{\det(M)} = \frac{18}{35}$$

Solve for slope m :

$$M_m = \begin{bmatrix} 9 & 18 \\ 29 & 56 \end{bmatrix} \quad \det(M_m) = 9(56) - 9(18) = 224 - 162 = 62$$

$$\text{apply Cramer's rule: } m = \frac{\det(M_m)}{\det(M)} = \frac{62}{35}$$

$$\text{Optimal least squares line: } \hat{y} = \frac{62}{35}x + \frac{18}{35}$$

b) Use the normal equations to find the same solution and verify it is the same as part a).

$$\theta^* = (X^T X)^{-1} \cdot X^T y \quad D = \{(0, 1), (2, 3), (3, 6), (4, 8)\}$$

$$\hat{y} = mx + b$$

$$\sum y_i = m \sum x_i + 4b$$

$$\text{parameter vector: } \theta = \begin{bmatrix} b \\ m \end{bmatrix}$$

$$\sum x_i y_i = m \sum x_i^2 + b \sum x_i$$

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \quad y = \begin{bmatrix} 1 \\ 3 \\ 6 \\ 8 \end{bmatrix}$$

$$X\theta = \begin{bmatrix} b \\ 2m + b \\ 3m + b \\ 4m + b \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix}$$

$$\text{recall } \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad X^T X^{-1} = \begin{bmatrix} 4 & 9 \\ 9 & 29 \end{bmatrix}^{-1} = \frac{1}{4(29) - 9(9)} \begin{bmatrix} 29 & -9 \\ -9 & 4 \end{bmatrix} = \frac{1}{116 - 81} \begin{bmatrix} 29 & -9 \\ -9 & 4 \end{bmatrix} = \frac{1}{35} \begin{bmatrix} 29 & -9 \\ -9 & 4 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 18 \\ 56 \end{bmatrix}$$

$$\theta^* = \frac{1}{35} \begin{bmatrix} 29 & -9 \\ -9 & 4 \end{bmatrix} \cdot \begin{bmatrix} 18 \\ 56 \end{bmatrix} = \frac{1}{35} \begin{bmatrix} 29(18) - 9(56) \\ -9(18) + 4(56) \end{bmatrix} = \frac{1}{35} \begin{bmatrix} 522 - 504 \\ -162 + 224 \end{bmatrix} = \frac{1}{35} \begin{bmatrix} 18 \\ 62 \end{bmatrix}$$

$$\theta^* = \begin{bmatrix} 18/35 \\ 62/35 \end{bmatrix} \leftarrow \text{same as Cramer's rule solution!}$$